Data science mini-project assignment

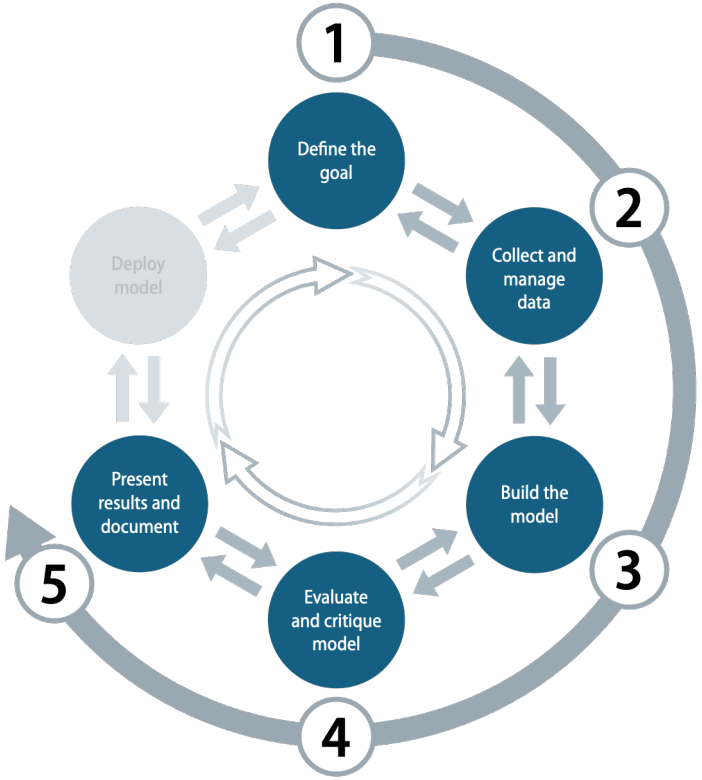
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Data science mini-project

Idea and motivation

- To **experience the life of a data scientist** on a small scale, where you will get some hands-on experience of the practice of:
  - Identifying a dataset
  - Setting up some hypotheses about the dataset that you would like to explore
  - Prepare the dataset for analysis
  - Selecting appropriate analysis method(s)
  - Train a model (models) to perform the analysis
  - Evaluate and communicate the results obtained

Complete a full data science project lifecycle
(except deployment)



1. Identify interesting questions, goals, or hypotheses
2. Choose data sets that most likely holds the answers, and perform the necessary data management
3. Build models that can be used to answer your questions
4. Validate the models with respect to suitable evaluation criteria
5. Analyze your results in terms of how your questions have been answered

   Present your work using text and visualizations

Formalities

Formalities

- 4.5 ECTS
- Individual, project-like assignment
- Grades: Pass/Fail (G/U)
- Grading is based on:
  1) **Submitted project** work in the form of well-documented code, e.g., *Jupyter Notebook* and *Docstrings* for Python
  2) **Oral presentation of the work before the entire class**, where the extension and depth of your understanding of the conducted work are assessed

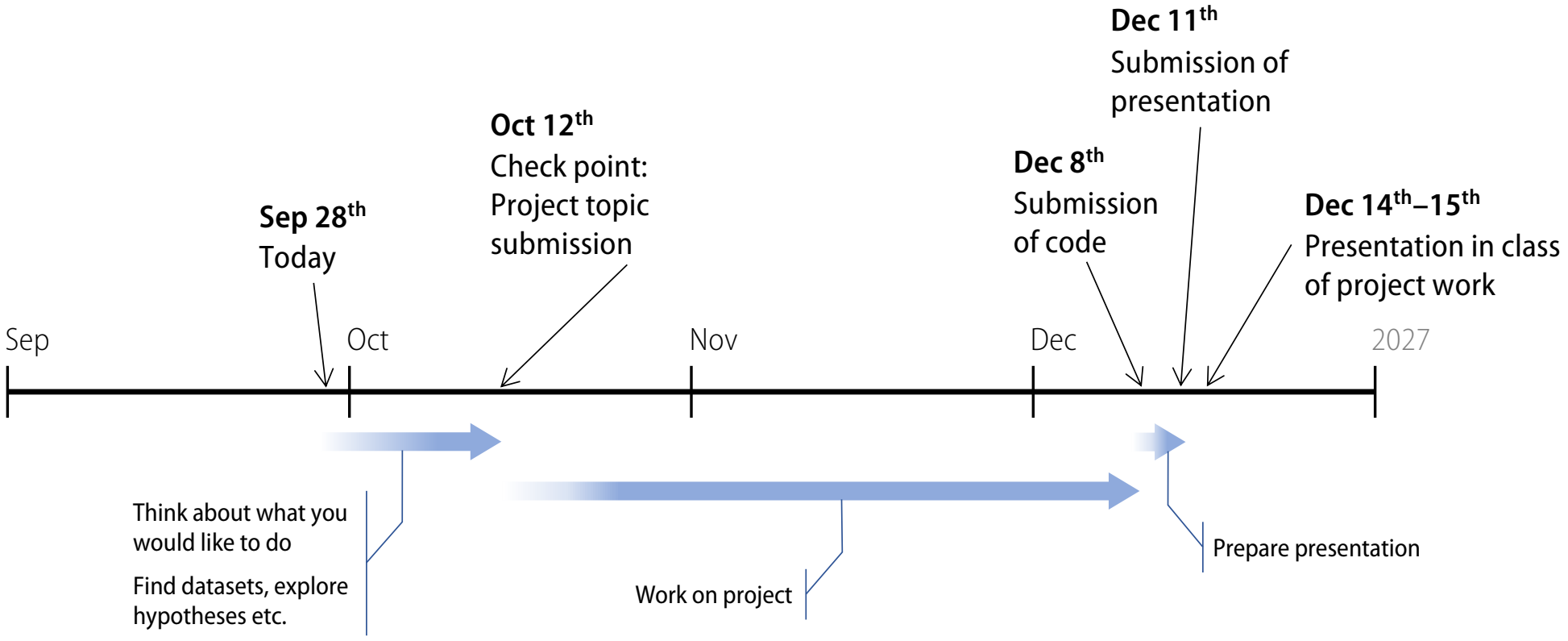
> **Note**: You should *not* write and submit a report describing your mini-project work

Time plan

Overall time plan

- **Introduction to the assignment**: The DS mini-project assignment is introduced in the lecture on September 28th (today)
- **Formulate project topic**: September 28th–October 12th (two weeks): Think about and formulate the topic for the DS mini-project, including identifying a suitable and accessible dataset
- **Work on project**: October 12th–December 8th (eight weeks): Work on the DS mini-project, addressing the first five steps in the DS project lifecycle while considering the assessment criteria
- **Prepare a project presentation**: December 8th–11th (four days): Prepare a presentation of your DS mini-project work, considering the assessment criteria
- **Present project**: December 14th–15th (either of the two days): Present and discuss your DS mini-project work in class

Timeline


Dec 11th
Submission of presentation
Oct 12th
Check point: Project topic submission
Dec 8th
Submission of code
Dec 14th–15th
Presentation in class of project work
Sep 28th
Today
Sep
Oct
Nov
Dec
2027
Think about what you would like to do
Find datasets, explore hypotheses etc.
Work on project
Prepare presentation

Project topic submission

- **No later than October 12th**, you should have submitted your mini-project topic through Canvas
- This is just **a check point**:
  - We will **not assess your submitted topic** from the perspective of how feasible the proposed mini-project is, i.e., you will not be given a *"this looks good"*, *"this is not enough"*, or similar
  - This is only to **make sure you're on track time-wise** so far, that you're in a position to start your mini-project work (i.e., you have a clear problem/question in mind, that you have identified a suitable and accessible dataset etc.)

Project topic submission (2)

- Submit a short summary of your chosen mini-project topic (maximum 1 page of text):
  - A (tentative) **heading** of your project, e.g., *"Predicting X using data Y"*
  - A **brief motivation** of the chosen topic, i.e., why is it of interest to you and also why it would serve as an interesting data science mini-project
  - **Problem/question** formulation, i.e., what you more specifically will try to solve/address
  - **Identified datasets** that will enable you to solve/answer your problem/question
  - If applicable, **identified previous work** that have solved/addressed your problem/question, including links/references

Choice of topic

Try to look ahead and identify a topic that you could expand upon in upcoming project and theses courses

- Choose a topic to investigate that is **of interest to you**
- You need to consider the following:
  - You will have to **complete a full cycle** of the data science project lifecycle (excluding deployment), i.e., you cannot pick a topic that just involves, e.g., data management and exploratory data analysis
  - We expect **some depth in the investigation** you carry out within the scope of your mini-project, which means that your topic – your problem/question and corresponding dataset – must allow you to **explore and expand the investigation beyond a superficial level**
- In the above sense, your problem/question and dataset must be **non-trivial**
- If you pick a previously well-investigated problem/question or dataset, we expect that you at least **try to find something new**, which was not obvious from the results of previous work, e.g., looking at the problem from a different angle, addressing a new sub-question, or using a new method of analysis

Code submission

- **No later than December 8th**, you should have submitted the **well-documented code** for your mini project through Canvas:
  - "Well-documented" means that the acceptable baseline is that you use the built-in features of the language in question (typically Python) to **comment and document your code** according to established practice
  - Much better is for you to **document your work** in the form of a "diary", where you describe your journey from start to finish, where each step ("entry") is accompanied with motivations for why it's pursued and comments on the achieved results, and where the **commented code is complemented with text and visuals**
  - For this, please consider using Jupyter Notebooks or similar

Presentation submission

- **No later than December 11th**, you should have submitted your **presentation materials**, which you will then deliver on-campus before the entire class at the designated time (on December 14th or 15th)
- In the oral presentation of the DS mini-project assignment, the **extension and depth of your understanding of the conducted work** are assessed:
  - *How and to what extent have you answered your question (solved your problem)?*
  - It's the **chain of reasoning that is important**: It's vital that you do not just describe *what* you did, but also explain *why* you did it (and not something else), as well as draw *conclusions and implications* from your results
- Make sure to apply what you've learned in the course regarding **correct use of tables, different types of graphs and plots, use of colors** etc.

Examination

Examination

- The presentations of the DS mini-projects will be conducted on-campus before the entire class, as **individual examinations**
- The format is a **15-minute presentation** (hard limit) by the student followed by **5 minutes of questions and discussion**
- Ordinary examination:
  - **December 14th** *or* **December 15th**
  - A 20-minute time slot should be booked in Canvas **no later than December 11th** (https://his.instructure.com/calendar#view_name=week&view_start=2026-12-14)
- Re-examination on **January 14th, 2027** (submissions on January 11th and 13th), same format and procedure as for the ordinary examination

Assessment

Overall basis for assessment

- **Level of ambition**:
  - The ability to **explore and expand a problem** from the trivial, superficial, or readily apparent, into a complex, in-depth, and novel investigation
- **Level of technical proficiency**:
  - The **adequacy and implementation of the components** of your investigation
  - It must be clear **how each step follows on the previous one**, what **alternative paths** from question to answer you tried and why they were discarded, and how you **decided on the selected path**
- **Level of scientific maturity**:
  - The ability to not only **describe what you have done**, but also **argue for why** you have used the chosen data, approaches, and methods, as opposed to potential alternatives, and **discuss the implications** of your results, as well as **reflect on your own work**, what you could have done differently, future work, ethical and societal aspects etc.

Supervision

Supervision

- Supervision is scheduled on six dates, totaling 18 hours (54 slots):
  - **October 22nd**, 9–12,
    November 5th, 9–12, **November 23rd**, 9–12, **November 30th**, 9–12,
    December 3rd, 9–12, and **December 8th**, 9–12
- Please note that these are **individual 20-minute time slots**, so not the entire class:
  - Takes place **on Zoom** (https://his-se.zoom.us/my/goranfalkman)
  - You book 20-minute time slots **on the supplied Google Spreadsheet** (https://docs.google.com/spreadsheets/d/10jXrEjcS4WJVpVQgqTG-lvm4E6h7DGTWlwZbJHCT1yg/edit?gid=0#gid=0)
  - **Only one slot can be booked in advance!**

Further help

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- For inspiration on a suitable topic for the mini-project, we have gathered **links to general project ideas as well as pages with interesting datasets** (https://his.instructure.com/courses/9907/pages/data-science-mini-project-project-ideas-and-datasets)
- To help you avoid the most common problems and pitfalls, we have compiled **general tips and advice** for the DS mini-project assignment from the experiences of previous years' mini-projects (https://his.instructure.com/courses/9907/pages/data-science-mini-project-tips-and-advice)
- See also *Introduction and overview – Data science mini-project* on Canvas: https://his.instructure.com/courses/9907/modules/items/240482

Questions?